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Dear Science Corps selection committee

I am writing this letter of recommendation in strong support of Ms. Sabrina Shirazi's application for the Science Corps. Sabrina is an exceptional graduate student finishing her PhD in the Ecology and Evolution program at the University of California, Santa Cruz. I supervised Sabrina for three years during her undergraduate internships at Smithsonian's Center for Conservation Genomics (CCG) and the Dept. of Anthropology at the National Museum of Natural History. I worked closely with Sabrina on a daily basis as she helped me complete my dissertation research integrating archaeological and genomic data. During her PhD, I have continued to work with Sabrina on a couple of genetics projects that were published and/or in review.

I first met Sabrina while I was looking for an undergraduate intern interested in anthropology and genetics. Sabrina was an exceptional undergraduate at the University of Maryland, where she came highly recommended by two of her anthropology professors. Sabrina completed all requirements for a BA in Anthropology and additionally created her own transdisciplinary major in Hominin Evolution; she graduated with double majors in Spring 2016. She excelled in all course work in the biological sciences including genetics and a very rigorous graduate course in molecular evolution. Along with a full course load, Sabrina has held two simultaneous internships at CCG and the National Museum of Health and Medicine where she gained experience working with human osteological collections.

CCG conducts research in molecular systematics, conservation genetics and evolutionary questions in a diverse set of organisms. In addition to working in the lab, Sabrina actively participated in many lab group activities including weekly journal club, public outreach at the Smithsonian's Autumn Conservation Festival and presented at the SCBI student research symposium. The Smithsonian is committed to using research to engage the public and Sabrina has been a great addition to outreach efforts. She is excellent at explaining very complex topics to diverse audiences and has impressed post-docs, graduate students and principal investigators with her commitment to outreach and engagement.

I do not typically mention this given gender bias in recommendation letters but given the requested information by Science Corps, Sabrina was a tremendously hard worker in her internship at CCG, often staying late into the evenings and completing lab work over the weekends. She is committed to completing any tasks that she initiates. Sabrina rapidly acquired lab skills including ancient and modern DNA extraction, library preparation, DNA capture, quantification, as well as Sanger and high-throughput sequencing. Sabrina is a quick learner and can immediately apply her new knowledge to provide innovative solutions to difficult problems. Sabrina volunteered on numerous occasions to test new lab equipment or develop new ways of using existing equipment. She was then able to successfully teach these methods/share this info with others in the lab as well as use them to further her own research

goals. She is inquisitive, engaged and enthusiastic, and other members of the lab appreciate her thoughtful work and attention to detail.

Besides technical skills, Sabrina also understands how to develop and conduct innovative research. She knows how to identify methods and resources appropriate to answer her questions and actively seeks out collaboration and assistance for problems that she cannot solve on her own. Sabrina has proactively worked to develop data analysis skills; during the summer of 2014 she took an introductory Unix/Linux workshop at the University of Maryland. She was also the first undergraduate to participate in the Programming for Evolutionary Biology workshop at the University of Leipzig, which is designed for senior graduate students and post-docs. She has used these skills to conduct analyses of nanopore sequence data using existing tools and continued developing bioinformatics skills during her PhD.

In our conversations about her career trajectory, she has expressed interest in working at a museum or running a program or initiative similar to Science Corps. She will excel at any of these endeavors. Sabrina has overcome a number of obstacles as woman of Persian descent. Her extended family has not always been supportive of her goal to pursue higher education, let alone a Ph.D. in evolutionary biology. She has mental and physical toughness having overcome a number of personal challenges in life. With her background in anthropology, ecology and evolution, I expect Sabrina will be an exceptional interdisciplinary researcher and teacher, as well as a strong mentor to students who have faced similar obstacles to their academic achievements. As one of the most driven and enthusiastic undergraduates that I have encountered, she has sought and obtained opportunities that have prepared her for a career in in science engagement and outreach.

Sabrina is not only an excellent researcher, mentor and colleague but she is a pleasure to spend time with and is willing to help her peers and supervisors with any task. In sum, I strongly recommend Sabrina and I urge you to consider her application favorably for Science Corps

Sincerely,

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